

Ho-Chien (Eric) Huang

Digital IC Design | EDA Scripting & AI-Assisted Workflows | DFT Background

Kaohsiung | +886 907-493-206 | hochienhuang1@gmail.com | linkedin.com/in/erichuang-tw | Available Dec 2026 (post-military service)

Summary

Dual master's graduate (EE, NCHU; CE, University of Cincinnati) with hands-on experience building TCL automation across the full RTL-to-GDSII EDA flow, plus Cadence Virtuoso layout experience. Two years applying deep learning / ML methods to industrial anomaly detection, and a structural test background (scan insertion, ATPG, STUMPS-based LBIST) reaching 99.78% transition fault coverage at 100 MHz. Targeting hardware design automation, EDA tooling, and AI-assisted design workflow roles.

Education

University of Cincinnati (UC) | Cincinnati, OH, USA

M.E. in Computer Engineering (Dual-Degree Program) | GPA 3.44/4.0 | Focus: VLSI Design / DFT | Aug 2025 – May 2026

National Chung Hsing University (NCHU) | Taichung, Taiwan

M.S. in Electrical Engineering | GPA 4.18/4.3 | Focus: Deep Learning / Anomaly Detection | Feb 2024 – May 2026

National Yunlin University of Science and Technology (NYUST) | Yunlin, Taiwan

B.E. in Interdisciplinary Studies (Minor: Mechanical Engineering) | GPA 3.83/4.00 | Sep 2018 – June 2023

- Phi Tau Phi Scholastic Honor Society, Honorary Member, 2023

Projects

MEng Capstone — DFT Implementation of STUMPS Architecture for 8-Bits Configurable FSMs Sequence Detector

University of Cincinnati | Advisor: Prof. Wen-Ben Jone (VLSI testing and BIST) | Oct 2025 – Apr 2026

- Designed an 8-bit Configurable FSM (CFSM) based on LUT architecture within a 310×310 μm area constraint, implementing 4,627 flip-flops at 100 MHz with 79.59% core area utilization.
- Inserted 10 parallel scan chains and ran ATPG with TetraMAX, achieving 99.78% transition fault coverage; implemented a STUMPS-based LBIST controller (PRPG, MISR, BIST FSM) and verified the golden MISR signature under 100 pseudo-random test patterns.
- Completed CTS optimization and RC extraction, confirmed timing closure via post-layout simulation, and passed DRC and LVS physical verification; added an output flip-flop to filter glitches.
- Automated the complete RTL-to-GDSII flow with custom TCL scripts across Synopsys Design Compiler (synthesis), IC Compiler II (P&R, CTS), and TetraMAX (ATPG), removing manual hand-offs between EDA tool stages; earlier prototyped on Cadence (Genus, Innovus, Xcelium) in an Intro to VLSI course project.

Master's Thesis — An Unsupervised Anomaly Detection System Based on Deep Reconstruction Feature Map Difference and Normalizing Flows

National Chung Hsing University | Advisor: Prof. Chia-Feng Juang (IEEE Fellow) | Feb 2024 – Aug 2025

- Combined a knowledge-distilled backbone with a reconstruction-based feature extractor so a single normalizing flow stack could handle multi-scale fusion.
- Reached 93.14% image-level AUROC on MVTec LOCO with a 17M parameter model, exceeding published baselines that use up to 16x parameters.
- Achieved 98.10% AUROC on a private industrial inspection dataset, exceeding all tested SOTA baselines.
- Measured 5.64 ms inference latency (176 FPS on an RTX 3080Ti) for single image on the MVTec LOCO benchmark.

Low-Power MNIST AI Accelerator — Hardware-Software Co-Design

University of Cincinnati | Course: Neuromorphic Computing for AI | Team of 3 (Hardware Mapping) | Feb 2026 – Apr 2026

- Built a self-checking C++ testbench with a golden model, reaching 0 mismatches across the test set in C/RTL Co-Simulation.
- Resolved a 138% DSP over-utilization (304 of 220 slices) by restructuring the MAC mapping with the software team, bringing it to 87.2%.
- Calibrated the requantization right-shift under an 8-bit multiplier constraint to minimize INT8 quantization error.
- Designed a 14-MAC parallel datapath and FSM-based BRAM controller in C++ with AMD Vitis HLS for a Zynq XC7Z020 FPGA board, and delivered it as an IP core.

Full On-Chip Low Dropout Voltage Regulator — Analog Design

University of Cincinnati | Course: Analog Circuits for CMOS Tech | Team of 3 (Layout & Circuit Debugging) | Feb 2026 – Apr 2026

- Designed a capacitor-less on-chip LDO in Cadence Virtuoso, completing full custom layout with device placement, routing, and DRC verification; characterized loop stability through gain-phase margin trade-off analysis (80 dB DC gain).

Work Experience

Software Development Intern — AOI Anomaly Detection System Development

Eyesontec Inc. | New Taipei City, Taiwan | Aug 2024 – Aug 2025

- Raised deep-learning AOI defect detection AUROC from 92.0% to 99.7% on production line data.
- Cut model training time by 90% and 50% inference time.
- Integrated Python deep-learning models with a C++ interface using pybind11 and a cross-platform dynamic-library approach, working around a legacy compiler that could not be upgraded.

Skills

Programming: Python (PyTorch, NumPy, pandas), C++, Verilog, TCL, Linux, Git (GitHub)

EDA Tools: Synopsys Design Compiler, IC Compiler II, TetraMAX; Cadence Virtuoso, Genus, Innovus, Xcelium; AMD Vivado, Vitis HLS

Design Flow: RTL-to-GDSII (synthesis, P&R, CTS, timing closure), post-layout simulation, DRC / LVS

Test & DFT: Scan insertion, ATPG, fault coverage analysis, STUMPS-based LBIST, golden-model testbench verification, C/RTL co-simulation

Languages: Mandarin (native), English (working proficiency), German (B1)